

3.1.1 Applied anatomy and physiology

Students should develop knowledge and understanding of the changes within the body systems prior to exercise, during exercise of differing intensities and during recovery.

Students should be able to interpret data and graphs relating to changes within the musculo-skeletal, cardio-respiratory and neuro-muscular systems and the use of energy systems during different types of physical activity and sport, and the recovery process.

3.1.1.1 Cardio-respiratory system

Students should understand the relationship between the cardiovascular and respiratory systems and how changes within these systems prior to exercise, during exercise of differing intensities and during recovery allow the body to meet the demands of exercise. They should also understand how taking part in physical activity and sport, as part of a healthy lifestyle, can have a positive effect on these systems.

3.1.1.2 Cardiovascular system

Content	Additional information
Understanding of the impact of physical activity and sport on the health and fitness of the individual.	Health (heart disease, high blood pressure, effects of cholesterol, stroke). Fitness (cardiac output – trained and untrained individuals, maximal and sub-maximal exercise).
The hormonal, neural and chemical regulation of responses during physical activity and sport.	Anticipatory rise. Redistribution of blood (vascular shunting vasoconstriction, vasodilation). Cardiac conduction system. Sympathetic and parasympathetic. Carbon dioxide.
Receptors involved in regulation of responses during physical activity.	Chemoreceptor, proprioceptor, baroreceptor.

Content	Additional information
Transportation of oxygen.	Haemoglobin. Myoglobin. Oxyhaemoglobin disassociation curve. Bohr shift.
Venous return.	Mechanisms. Relationship with blood pressure (systolic, diastolic).
Starling's law of the heart.	
Cardiovascular drift.	
Arterio-venous oxygen difference (A-VO ₂ diff).	Variations in response to an exercise session. Variations between trained and untrained individuals. Adaptations to body systems resulting in training effect.

3.1.1.3 Respiratory system

Content	Additional information
Understanding of lung volumes and the impact of and on physical activity and sport.	Residual volume. Expiratory reserve volume. Inspiratory reserve volume. Tidal volume. Minute Ventilation.
Gas exchange systems at alveoli and muscles.	Oxygen and carbon dioxide. Principles of diffusion and partial pressures.
The neural and chemical regulation of pulmonary ventilation during physical activity and sport.	Sympathetic and parasympathetic. Carbon dioxide.
Receptors involved in regulation of pulmonary ventilation during physical activity.	Chemoreceptor, proprioceptor, baroreceptor.
Impact of poor lifestyle choices on the respiratory system.	Smoking. Oxygen transport.

3.1.1.4 Neuromuscular system

Students should understand the relationship between the nervous and muscular systems and how changes within these systems prior to exercise, during exercise of differing intensities and during recovery allow the body to meet the demands of exercise.

Content	Additional information
Characteristics and functions of different muscle fibre types for a variety of sporting activities.	Slow twitch (type I). Fast glycolytic (type IIx). Fast oxidative glycolytic (type IIa).
Nervous system.	Sympathetic and parasympathetic.
Role of proprioceptors in PNF.	Muscle spindles. Golgi tendon organ.
The recruitment of muscle fibres.	Motor units. Spatial summation. Wave summation. All or none law. Tetanic.

3.1.1.5 The musculo-skeletal system and analysis of movement in physical activities

Students should understand the relationship between the muscular and skeletal systems to meet the demands of exercise. Students should be able to apply their knowledge and understanding to specific sporting actions and movement in a range of physical activities.

Content	Additional information
Joint actions in the sagittal plane/transverse axis.	Shoulder and hip (flexion, extension and hyperextension). Elbow and knee (flexion and extension). Ankle (plantar flexion and dorsi flexion).
Joint actions in the frontal plane/sagittal axis.	Shoulder and hip (adduction and abduction).
Joint actions in the transverse plane/longitudinal axis.	Shoulder and hip (horizontal abduction and adduction).
Types of joint, articulating bones, main agonists and antagonists, types of muscle contraction.	Isotonic (concentric and eccentric) and Isometric.

3.1.1.6 Energy systems

Students should develop knowledge and understanding of energy systems prior to exercise, during exercise of differing intensities and during recovery.

Content	Additional information
Energy transfer in the body.	Aerobic energy system (glycolosis, kreb/citric acid cycle, beta oxidation, electron transport chain). Anaerobic energy systems (ATP-PC system, anaerobic glycolytic system).
Energy continuum of physical activity.	Consideration for physical activity and sport of different intensities and durations. Differences in ATP generation between fast and slow twitch muscle fibre.
Energy transfer during short duration/high intensity exercise.	Anaerobic energy system. ATP-PC system. Anaerobic glycolytic system (lactate accumulation, lactate threshold, OBLA, lactate producing capacity and sprint/power performance).
Energy transfer during long duration/lower intensity exercise.	Aerobic energy system. Oxygen consumption during exercise (maximal and submaximal oxygen deficit). Oxygen consumption during recovery (excess post-exercise oxygen consumption EPOC).
Factors affecting VO_2 max/aerobic power.	
Measurements of energy expenditure.	Indirect calorimetry. Lactate sampling. VO_2 max test. Respiratory exchange ratio (RER).
Impact of specialist training methods on energy systems.	Altitude training. High Intensity Interval Training (HIIT). Plyometrics. Speed Agility Quickness.